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Bayesian Inference for Matrix-Variate Gaussian Distributions: Theory, Computation, and Applications

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Abstract:

We develop a fully Bayesian framework for matrix-variate Gaussian distributions with Kronecker-structured covariance $\Psi \otimes \Sigma$, including conjugate priors, Gibbs sampling full conditionals, and sparsity-inducing extensions for high dimensions. The inherent scaling identifiability issue $(\Psi \otimes \Sigma) = (c\Psi) \otimes (c^{-1}\Sigma)$ is addressed via the constraint $\text{tr}(\Sigma) = p$, though simulations reveal asymmetry when $p \neq q$. Increasing sample size from $n = 30$ to $n = 100$ reduces relative Frobenius error by 55% for dimensions $p, q \leq 10$.

Keywords: Bayesian inference, Gibbs sampling, Kronecker covariance, Matrix-variate Gaussian.

Mathematics Subject Classification (2010): 62F15, 62H12, 62J07.

1 Introduction

Random matrices arise across neuroimaging (subject $\times$ region), genomics (genes $\times$ time), finance (assets $\times$ days), and environmental monitoring (locations $\times$ time). The matrix-variate Gaus-

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sian distribution captures both row-wise and column-wise dependencies via a separable Kronecker product covariance: $\text{vec}(X) \sim \mathcal{N}_{pq}(\text{vec}(M), \Psi \otimes \Sigma)$, with density

$$f(X \mid M, \Sigma, \Psi) = \frac{\exp\left(-\frac{1}{2} \text{tr} \left[\Sigma^{-1} (X - M) \Psi^{-1} (X - M)^\top \right]\right)}{(2\pi)^{pq/2} |\Sigma|^{q/2} |\Psi|^{p/2}}.$$

The Kronecker structure reduces parameters from $\mathcal{O}(p^2 q^2)$ to $\mathcal{O}(p^2 + q^2)$, essential when $pq \gg n$ [Stegle et al. \(2011\)](#). Recent advances include warped matrix-variate Gaussian processes for multi-step prediction [Takano and Jimbo \(2025\)](#), theoretical treatments of matrix-valued Gaussian processes [Diaz et al. \(2022\)](#), and foundational distributional properties [Barratt \(2018\)](#).

While frequentist MLE via the flip-flop algorithm [Dutilleul \(1999\)](#) provides only point estimates and fails when $p, q > n$, Bayesian methods address these via full posteriors and regularization. The conjugate prior framework was pioneered by [Dawid \(1981\)](#) and [Brown et al. \(1993\)](#), with extensions to deep matrix-variate Gaussian processes [Park et al. \(2018\)](#). For high dimensions, sparsity-inducing priors [Wang \(2012\)](#) and tensor decompositions [Petersen et al. \(2019\)](#) have proven effective.

This paper provides a unified Bayesian treatment of matrix-variate Gaussian models. Section 2 develops conjugate priors and Gibbs sampling. Section 3 extends to sparsity-inducing priors (Bayesian graphical lasso, spike-and-slab). Section 4 discusses variational Bayes for scalability and presents simulation studies examining sample size, dimensionality, and identifiability constraints. Finally, the discussion and suggestions for future work are presented.

2 Conjugate Bayesian Inference

Due to the exponential family nature of the matrix-variate Gaussian distribution, a conjugate prior exists. Following [Dawid \(1981\)](#) and [Brown et al. \(1993\)](#), we specify:

$$M \mid \Sigma, \Psi \sim \mathcal{MN}_{p \times q}(M_0, \Sigma_0, \Psi_0), \quad \Sigma^{-1} \sim \mathcal{W}_p(\nu_\Sigma, V_\Sigma), \quad \Psi^{-1} \sim \mathcal{W}_q(\nu_\Psi, V_\Psi), \quad (2.1)$$

where $\mathcal{W}_d(\nu, V)$ is the Wishart distribution with $\nu > d - 1$ degrees of freedom and scale V . Hyperparameters include prior mean M_0 , row/column scale matrices Σ_0, Ψ_0 , degrees of freedom ν_Σ, ν_Ψ , and Wishart scales V_Σ, V_Ψ , all specified by the analyst.

Conjugacy implies the posterior belongs to the same family. For i.i.d. observations $X_1, \dots, X_n \sim \mathcal{MN}_{p \times q}(M, \Sigma, \Psi)$, define $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and

$$S_\Sigma(M) = \sum_{i=1}^n (X_i - M) \Psi^{-1} (X_i - M)^\top, \quad S_\Psi(M) = \sum_{i=1}^n (X_i - M)^\top \Sigma^{-1} (X_i - M).$$

Although M appears in both $S_\Sigma(M)$ and $S_\Psi(M)$, preventing a simple factorization of the joint posterior, each parameter's full conditional remains conjugate. This naturally motivates a Gibbs sampling strategy.

The full conditionals for Gibbs sampling are given below.

Theorem 2.1. Under the prior specified in (2.1), the full conditional distributions are:

$$M \mid \Sigma, \Psi, X \sim \mathcal{MN}_{p \times q}(\widetilde{M}, \widetilde{\Sigma}, \widetilde{\Psi}), \quad (2.2)$$

$$\Sigma^{-1} \mid M, \Psi, X \sim \mathcal{W}_p \left(nq + \nu_\Sigma, \left[V_\Sigma^{-1} + \sum_{i=1}^n (X_i - M) \Psi^{-1} (X_i - M)^\top \right]^{-1} \right), \quad (2.3)$$

$$\Psi^{-1} \mid M, \Sigma, X \sim \mathcal{W}_q \left(np + \nu_\Psi, \left[V_\Psi^{-1} + \sum_{i=1}^n (X_i - M)^\top \Sigma^{-1} (X_i - M) \right]^{-1} \right), \quad (2.4)$$

where

$$\widetilde{\Sigma} = (\Sigma_0^{-1} + n\Sigma^{-1})^{-1}, \quad \widetilde{\Psi} = (\Psi_0^{-1} + n\Psi^{-1})^{-1}, \quad \widetilde{M} = \widetilde{\Sigma} \left(\Sigma_0^{-1} M_0 \widetilde{\Psi}^{-1} + \Sigma^{-1} \sum_{i=1}^n X_i \Psi^{-1} \right) \widetilde{\Psi}.$$

These conditional distributions yield a straightforward Gibbs sampling scheme: at each iteration, sample M from (2.3), then Σ^{-1} from (2.4), and finally Ψ^{-1} from (2.5). Due to the Gaussian-Wishart structure, convergence is typically fast.

Hyperparameter selection significantly impacts posterior inference. For weakly informative priors, we recommend:

- $M_0 = 0_{p \times q}$, $\Sigma_0 = cI_p$, $\Psi_0 = cI_q$ with a large c (e.g., $c = 100$);
- $\nu_\Sigma = p + 2$, $\nu_\Psi = q + 2$ (minimum for finite prior means), and $V_\Sigma = I_p$, $V_\Psi = I_q$.

Alternatively, empirical Bayes via marginal maximum likelihood (EM algorithm) can estimate hyperparameters.

Posterior summaries are obtained directly from the Gibbs samples. For predicting a new observation X_{new} , the posterior predictive density is given by:

$$p(X_{\text{new}} \mid X) = \iint p(X_{\text{new}} \mid M, \Sigma, \Psi) p(M, \Sigma, \Psi \mid X) dM d\Sigma d\Psi,$$

which can be approximated by Monte Carlo averaging over the saved posterior samples. Under the conjugate prior, the posterior predictive distribution is matrix-variate t , allowing for closed-form predictive densities when desired.

3 Sparsity-Inducing Priors for High Dimensions

The conjugate Wishart prior alone is insufficient for regularization when p or q exceeds n . We therefore introduce sparsity-inducing priors on the precision matrices $\Omega_\Sigma = \Sigma^{-1}$ and $\Omega_\Psi = \Psi^{-1}$. Following Wang (2012) and Petersen et al. (2019), we place Laplace priors on the off-diagonal elements of the precision matrices:

$$p(\Omega_\Sigma \mid \lambda_\Sigma) \propto \prod_{i < j} \frac{\lambda_\Sigma}{2} \exp(-\lambda_\Sigma |\omega_{ij, \Sigma}|) \cdot \prod_{i=1}^p \frac{\lambda_\Sigma}{2} \exp\left(-\frac{\lambda_\Sigma}{2} \omega_{ii, \Sigma}\right) \mathbf{1}_{\Omega_\Sigma \in \mathcal{P}_p^+},$$

$$p(\Omega_\Psi \mid \lambda_\Psi) \propto \prod_{i < j} \frac{\lambda_\Psi}{2} \exp(-\lambda_\Psi |\omega_{ij, \Psi}|) \cdot \prod_{i=1}^q \frac{\lambda_\Psi}{2} \exp\left(-\frac{\lambda_\Psi}{2} \omega_{ii, \Psi}\right) \mathbf{1}_{\Omega_\Psi \in \mathcal{P}_q^+},$$

where $\mathcal{P}_d^+$ denotes the set of $d \times d$ positive definite matrices. The tuning parameters $\lambda_\Sigma, \lambda_\Psi$ control the degree of sparsity. For full Bayesian inference, we assign gamma hyperpriors: $\lambda_\Sigma \sim \text{Gamma}(a_\lambda, b_\lambda)$, and similarly for λ_Ψ .

Posterior sampling employs block Gibbs updates with double Metropolis-Hastings or the Bayesian graphical lasso algorithm Wang (2012), which augments the model with latent scale parameters to represent the Laplace distribution as a scale mixture of normals:

$$\frac{\lambda}{2} e^{-\lambda|\beta|} = \int_0^\infty \frac{1}{\sqrt{2\pi\tau^2}} e^{-\beta^2/(2\tau^2)} \frac{\lambda^2}{2} e^{-\lambda^2\tau^2/2} d\tau^2.$$

For explicit model selection, spike-and-slab priors offer a more flexible approach. For $i < j$,

$$\omega_{ij, \Sigma} \mid \gamma_{ij, \Sigma} \sim (1 - \gamma_{ij, \Sigma})\delta_0 + \gamma_{ij, \Sigma}\mathcal{N}(0, \tau_\Sigma^2),$$

$$\gamma_{ij, \Sigma} \mid \pi_\Sigma \sim \text{Bernoulli}(\pi_\Sigma), \quad \pi_\Sigma \sim \text{Beta}(a_\pi, b_\pi),$$

where δ_0 is a point mass at zero and τ_Σ^2 is chosen large. Diagonal elements are constrained positive, typically with exponential priors. This prior enables posterior inference on which partial correlations are non-zero, providing a Bayesian analogue of covariance selection, though the positive definiteness constraint must be enforced separately.

For many applications, the covariance exhibits low-rank plus sparse structure (e.g., factor models). We decompose:

$$\Sigma = \Sigma_L + \Sigma_S + \sigma^2 I_p, \quad \Psi = \Psi_L + \Psi_S + \tau^2 I_q,$$

with Σ_L low-rank, Σ_S sparse, and $\sigma^2 I_p$ diagonal. Bayesian inference employs multiplicative gamma process priors for the low-rank component Bhattacharya and Dunson (2015) and horseshoe priors for the sparse component.

4 Computational Strategies and Simulation Studies

Algorithm 1 summarizes the Gibbs sampler for the conjugate model.

Algorithm 1. Gibbs Sampler for Matrix-Variate Gaussian Model

- Initialize $M^{(0)} = \bar{X}$, $\Sigma^{(0)} = I_p$, $\Psi^{(0)} = I_q$.
- For $t = 1, \dots, T$:
 1. Sample $M^{(t)} \mid \Sigma^{(t-1)}, \Psi^{(t-1)}, X$ from (2.3).
 2. Compute $S_\Sigma = \sum_{i=1}^n (X_i - M^{(t)})(\Psi^{(t-1)})^{-1}(X_i - M^{(t)})^\top$.
 3. Sample $\Sigma^{(t)} \sim \mathcal{W}_p^{-1}(nq + \nu_\Sigma, (V_\Sigma^{-1} + S_\Sigma)^{-1})$.
 4. Compute $S_\Psi = \sum_{i=1}^n (X_i - M^{(t)})^\top (\Sigma^{(t)})^{-1} (X_i - M^{(t)})$.
 5. Sample $\Psi^{(t)} \sim \mathcal{W}_q^{-1}(np + \nu_\Psi, (V_\Psi^{-1} + S_\Psi)^{-1})$.
- Discard first B iterations as burn-in.

Gibbs sampling becomes slow for large p and q (e.g., $p, q > 100$). Variational Bayes (VB) provides a deterministic approximation. We posit a mean-field factorization:

$$q(M, \Sigma, \Psi) = q_M(M) q_\Sigma(\Sigma) q_\Psi(\Psi).$$

The VB algorithm iteratively updates each factor by minimizing the Kullback-Leibler divergence $\text{KL}(q \| p(\cdot \mid X))$, yielding:

$$\begin{aligned} \log q_M(M) &= \mathbb{E}_{q_\Sigma q_\Psi} [\log p(X \mid M, \Sigma, \Psi) + \log p(M \mid \Sigma, \Psi)] + \text{const}, \\ \log q_\Sigma(\Sigma^{-1}) &= \mathbb{E}_{q_M q_\Psi} [\log p(X \mid M, \Sigma, \Psi) + \log p(\Sigma^{-1})] + \text{const}, \end{aligned}$$

and similarly for Ψ . The updates become:

$$q_M(M) = \mathcal{MN}(\widehat{M}, \widehat{\Sigma}, \widehat{\Psi}), \quad q_\Sigma(\Sigma^{-1}) = \mathcal{W}_p(\widehat{\nu}_\Sigma, \widehat{V}_\Sigma), \quad q_\Psi(\Psi^{-1}) = \mathcal{W}_q(\widehat{\nu}_\Psi, \widehat{V}_\Psi),$$

with parameters updated via coordinate ascent. VB converges in tens of iterations, each requiring $\mathcal{O}(p^3 + q^3)$ operations, making it suitable for moderate dimensions ($p, q \sim 200$).

We conducted simulation studies to evaluate the finite-sample performance of the proposed Bayesian Gibbs sampler for matrix-variate Gaussian estimation. Our primary objectives were to assess: (i) the effect of sample size on estimation accuracy, (ii) the impact of row and column dimensions, and (iii) the sensitivity to the identifiability constraint required for the Kronecker product decomposition.

Data were generated from the matrix-variate Gaussian distribution $\mathcal{MN}_{p \times q}(0, \Sigma, \Psi)$ with zero mean matrix to isolate covariance estimation. For all simulations, we used the weakly informative prior specification described in Section 2:

$$M_0 = 0_{p \times q}, \quad \Sigma_0 = 100I_p, \quad \Psi_0 = 100I_q, \quad \nu_\Sigma = p + 2, \quad V_\Sigma = I_p, \quad \nu_\Psi = q + 2, \quad V_\Psi = I_q.$$

The Gibbs sampler was run for 15,000 iterations with the first 5,000 discarded as burn-in. To resolve the scaling indeterminacy inherent in the Kronecker product ($\Psi \otimes \Sigma = c\Psi \otimes c^{-1}\Sigma$ for any $c > 0$), we imposed the identifiability constraint $\text{tr}(\Sigma) = p$, rescaling both covariance matrices after each iteration.

We examined five representative configurations:

- Small symmetric: $p = 5, q = 5, n = 30$;
- Asymmetric (rows < columns): $p = 5, q = 10, n = 30$;
- Asymmetric (rows > columns): $p = 10, q = 5, n = 30$;
- Moderate symmetric: $p = 10, q = 10, n = 30$;
- Large sample: $p = 10, q = 10, n = 100$.

For simplicity and to establish baseline performance, we set the true covariance matrices to identity:

$\Sigma = I_p$ and $\Psi = I_q$. Estimation accuracy was measured using the relative Frobenius norm error:

$$\text{Error}(\hat{\Sigma}) = \frac{\|\hat{\Sigma} - \Sigma\|_F}{\|\Sigma\|_F}, \quad \text{Error}(\hat{\Psi}) = \frac{\|\hat{\Psi} - \Psi\|_F}{\|\Psi\|_F},$$

where $\hat{\Sigma}$ and $\hat{\Psi}$ are posterior means, and $\|\cdot\|_F$ denotes the Frobenius norm.

Table 1 presents the estimation errors for all five configurations.

Table 1: Estimation errors after applying the identifiability constraint $\text{tr}(\Sigma)=p$

Configuration	p	q	n	Σ Error
Small symmetric	5	5	30	0.211
Asymmetric (rows < cols)	5	10	30	0.155
Asymmetric (rows > cols)	10	5	30	0.274
Moderate symmetric	10	10	30	0.212
Large sample	10	10	100	0.096

The sampler demonstrates desirable asymptotic behavior: increasing the sample size from $n = 30$ to $n = 100$ reduced the estimation error by 55% (from 0.212 to 0.096). All errors are below

0.28, indicating reliable performance for dimensions $p, q \leq 10$ with sample size $n \geq 30$. The small symmetric configuration ($p = q = 5, n = 30$) achieved an error of 0.211, while the moderate symmetric configuration ($p = q = 10, n = 30$) produced a nearly identical error of 0.212, suggesting that doubling the dimension from 5 to 10 does not substantially degrade performance when the sample size is fixed.

We observe mild asymmetry in the results: the error for the configuration with fewer rows than columns ($p = 5, q = 10$) is 0.155, while the error for the transposed configuration ($p = 10, q = 5$) is 0.274. Under perfect symmetry, these values should be identical, as the model treats rows and columns symmetrically.

This asymmetry suggests that the trace constraint $\text{tr}(\Sigma) = p$ does not fully resolve the Kronecker product scaling indeterminacy. The constraint fixes the scale of Σ but leaves the relative scaling between Σ and Ψ partially unidentified, particularly when $p \neq q$. As a result, the Gibbs sampler may experience different mixing behavior depending on which dimension is larger.

Alternative constraints may yield perfect symmetry. For instance, fixing $\Sigma_{11} = 1$ (i.e., setting the first diagonal element of Σ to unity) provides a direct scale anchor. Another approach is to impose $\|\Sigma\|_F = \sqrt{p}$, which normalizes the Frobenius norm. We leave a systematic comparison of identifiability constraints for future research.

Discussion and Conclusion

We developed a Bayesian framework for matrix-variate Gaussian distributions with Kronecker covariance. The trace constraint resolves scaling indeterminacy but introduces asymmetry when $p \neq q$ (errors 0.155 vs 0.274). Simulations show increasing n reduces error for $p, q \leq 10$. Future work includes non-Gaussian extensions, and comparison of alternative identifiability constraints.

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